

ZEROPAY DIGITAL SERVICES LTD

United Kingdom Subsidiary of BETONIQ WEST LTD (RC 1496603, Nigeria)

ChipLink — Technical White Paper

Fabless mature-node chip sourcing and design brokerage.

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Executive Summary

ChipLink is a Zeropay Digital Services Ltd venture (UK subsidiary of BETONIQ WEST LTD) — Fabless mature-node chip sourcing and design brokerage.

Key statistic: A handful of companies control the world's advanced semiconductor supply.

Leading-edge fabrication (below 7nm) requires ASML EUV lithography tooling, export-controlled to a small number of countries and costing \$10-20 billion+ per fab — but the mature-node tier (28nm and above) remains open to new entrants.

Problem Statement

Small and mid-sized hardware manufacturers, including African fintech and IoT makers, are frequently locked out of chip supply by minimum-order-quantity thresholds and broker markups, even for mature-node components that do not require cutting-edge fabrication.

Technical Architecture

- RFQ marketplace lets a buyer submit a chip specification and quantity; verified mature-node foundries and OSAT partners bid competitively and transparently.
- Order-pooling engine aggregates volume across multiple small buyers with compatible specifications to collectively reach a foundry's minimum-order-quantity threshold.
- Fabless design-support layer helps a buyer without in-house chip expertise translate a product requirement into a manufacturable specification.
- Explicit scope boundary: mature-node (28nm+) sourcing and OSAT packaging/testing only — no leading-edge fabrication.

Core Data Model

Entity	Purpose
Buyer	Hardware manufacturer submitting a chip specification and required volume.
ChipSpec	Technical specification: process node, package type, functional requirements.
FoundryPartner	Verified mature-node foundry or OSAT partner, capacity and lead time.
RFQ	Request-for-quote record with competing bids from foundry partners.
OrderPool	Aggregated volume from multiple buyers pooled to reach a foundry's minimum order quantity.

Security, Privacy & Compliance

- Buyer chip specifications treated as confidential; visible only to bidding foundry partners under a standard non-disclosure framework.
- Foundry partner verification process to guard against counterfeit or non-compliant component sourcing.
- Order-pooling logic ensures no single pooled buyer can see another pooled buyer's full specification or pricing terms.

Feasibility & Build Status

RFQ marketplace and order-pooling logic live on Hugging Face Spaces. This platform also directly serves Zeropay/BETONIQ WEST's own hardware supply chain need for PUF/ECC secure element components.

Market Opportunity

Mature-node semiconductor market alone exceeds \$60B annually. Target beachhead: African/Middle-Eastern fintech and IoT hardware makers currently priced out by MOQ thresholds.

Grant & Funder Fit

Fits industrial policy and supply-chain resilience funding: US CHIPS Act adjacent programs, EU Chips Act supply-chain diversification funds, India Semicon Mission, and African Union/AfDB regional value-chain development grants.

Funding Ask

\$200,000 grant to build RFQ marketplace + onboard first 10 verified mature-node foundry/OSAT partners, 12 months.